

Christian Otty

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EDUCATION

Binghamton University, State University of New York

Master of Science in Electrical and Computer Engineering

May 2026

Bachelor of Science in Computer Engineering

May 2025

Cumulative GPA: 3.62/4.00

SKILLS

Languages: C, C++, HTML, CSS, JavaScript, Python, Bash, PowerShell, SQL, MATLAB, Verilog

Software: Visual Studio, SOLIDWORKS, KiCad, Elasticsearch, Ansible, Fusion360, Xilinx ISE

Certifications: Elastic Certified Engineer

PROFESSIONAL EXPERIENCE

Systems & Infrastructure Engineer, New York Information Technology Services | Albany, NY

July 2025 – Present

- Managed Elasticsearch infrastructure serving multiple state agencies, gaining extensive hands-on experience with Linux system administration, server configuration, and automation workflows
- Led infrastructure modernization initiative by developing Ansible automation to transition over 200 servers from legacy Beats stack to modern Elastic Agent architecture, improving system efficiency and maintainability
- Reduced log ingestion by 30% (14 million logs per hour) through automated removal of redundant logging agents, improving query performance and data visualization for customers

Platform Automation Engineer, New York Information Technology Services | Albany, NY

May 2024 – August 2024

- Partnered with professional developers to perform server maintenance, automate system processes, and contribute to backend web development projects
- Employed Microsoft Razor Pages to integrate PowerShell scripts into the organization's website, creating a dynamic function to improve efficiency and facilitate the addition of future features
- Created PowerShell and SQL scripts to automate data collection and support server stability

PROJECTS

Custom Wireless Mouse, Personal Project

January 2026 – Present

- Engineered a custom BLE optical mouse, designing a PCB in KiCad around an nRF52840 SoC and PMW3389 sensor with a Li-ion power system (USB-C charging, protection circuitry) across 2 revisions
- Developed BLE HID firmware on Zephyr RTOS using the nRF Connect SDK, extending the nRF Desktop reference application to support motion, button, and scroll input
- Resolved SWD debugging and BLE limitations on an initial dev board by migrating to an nRF52840 dongle; currently redesigning around a bare SoC with custom power/charging circuitry to remove the dev-board dependency.

Homelab, Personal Project

July 2025 – Present

- Provisioned a Proxmox virtualization environment with Terraform and Ansible, then re-architected it into a 30+ container bare-metal Docker Compose deployment with fully externalized configuration
- Built a custom Caddy reverse-proxy image to support automated wildcard TLS issuance (ACME DNS-01), eliminating manual certificate work for every new service
- Applied a default-deny access-control layer with self-hosted Authelia (OIDC SSO), authenticating every service by default and documenting protocol-specific exceptions only where a client genuinely couldn't support redirect-based auth.

Portable Solar Power Station, Senior Design Project

August 2024 – May 2025

- Collaborated with a team of four to research, design, and build a portable solar power station sponsored by Avangrid, showcasing renewable energy applications for small-scale use cases
- Conducted research on battery cells and management systems to ensure a safe and reliable energy storage design
- Programmed an ESP32 to decode BMS UART data and render real-time battery statistics on an LCD with efficient refresh rates for clear and responsive monitoring

Accumulator Sub-Team, Society of Automotive Engineers (SAE)

September 2023 – June 2025

- Contributed to the assembly, wiring, and maintenance of the high-voltage battery system for the team's electric drive formula car, ensuring compliance with FSAE safety regulations
- Developed a PCB fuse board for individual battery modules, routing sense wires to the battery management system
- Programmed a C++ based diagnostic function for an STM32 microcontroller to demonstrate flash memory manipulation and data communication between technician systems